Tony Zhang

Kahlert School of Computing
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EDUCATION

B.S. Data Science, Minor in Robotics, University of Utah (expected) May 2027
Kahlert School of Computing · GPA 3.94 / 4.0

Exchange Student, Department of Computer Science and Technology, Tsinghua University Spring 2026
Beijing, China · Graduate coursework in machine learning systems

- Course project: benchmarked Flash Sparse Attention against Native Sparse Attention on an NVIDIA A10 GPU (Hardware/Software Co-designed AI Systems).

RESEARCH EXPERIENCE

Undergraduate Researcher — ARIA Lab, Robotics Center, University of Utah Jul 2025 – Present
Advisor: Dr. Daniel S. Brown

- Lead researcher on IBMARL (Imitation Bootstrapped Multi-Agent Reinforcement Learning), which extends IBRL to cooperative teams. Round-Robin Behavior Cloning lets one human supply demonstrations for a whole team; an ensemble of centralized critics chooses, per agent, between the imitation and RL actions; a gated imitation loss pulls each RL actor toward the imitation policy only where the critics prefer it, so the final policies run decentralized without the critics.
- On four cooperative VMAS benchmarks, IBMARL surpasses imitation learning 2.6–4.2× faster than demonstration-based baselines and exceeds the demonstrator itself on two tasks. With real human demonstrations it surpasses imitation learning more than twice as fast as the best baseline.
- Trained diffusion-based behavior cloning policies for SO-101 robot arms in PyTorch.

Resulted in a first-author workshop paper at IROS 2026 (Best Paper Finalist) and a first-author submission to AAMAS 2027.

NSF REU Researcher — FROST Lab, Brigham Young University Jun – Aug 2026
Advisor: Dr. Brady Moon

- Developed CONSORT, a centralized planner for multi-agent path planning with cooperative localization in GPS-denied environments. A primary agent must reach its goal in minimum time under a terminal bound on localization uncertainty and chance-constrained collision avoidance; support agents are routed only to hold the primary under that bound (the “escort problem”).
- Made the belief-space problem tractable by optimizing mean trajectories only and propagating covariance to evaluate constraints; paired joint search on a heading-aware hexagonal lattice with B-spline refinement under curvature, collision, and uncertainty constraints.
- On 50 randomized scenarios, CONSORT solved 80% where the strongest of four baselines solved 54%, at comparable path length. Monte Carlo validation in HoloOcean with an independent factor-graph estimator kept trajectories collision-free with terminal uncertainty within 6% of the planner’s prediction; this simulation study replaced field trials after a vehicle failure.
- Learned surface-mount assembly to help build the Central Utility Board (CUB), the lab’s custom PCB for its AUVs, and helped with vehicle bring-up; acknowledged in B. Washburn et al., “CougarTail & CUB: A General-Purpose Mast and Central Utility Board for Cylindrical Underwater Enclosures,” arXiv:2609.10230, 2026.

Resulted in a first-author submission to ICRA 2027. Julia implementation and benchmark scenarios released on the lab project page. Supported by NSF Award 2447307.

Undergraduate Researcher — Adler Lab, Mathematical Biology, University of Utah Jan 2024 – May 2025
Advisor: Dr. Fred Adler · Through the College of Science’s Science Research Initiative (SRI)

- Awarded a UROP grant and SRI summer funding to model the immune response to heterogeneous tumors: built 1D and 2D simulations of chemokine gradients (steady-state diffusion) and T-cell migration (von Mises directional sampling) in R and C++, and compared them with mouse-model data from the Reeves Lab at Huntsman Cancer Institute.
- Built ML pipelines in Python and R to score melanoma cell-track quality from quantitative phase imaging, with the Judson-Torres Lab at Huntsman Cancer Institute; presented as a poster at the Math4All Conference, 2024.

- Visualization of the T-cell migration simulation was nominated as one of the best projects in Scientific Visualization (Prof. Chris Johnson, Spring 2025) and presented to the class.

PUBLICATIONS

Submitted (under review)

1. **T. Zhang**, R. Bartholomew, K. Velasco, J. Mangelson, B. Moon. "CONSORT: Cooperative Navigation with Support-Optimized Routing and Trajectories for GPS-Denied Escort Missions." Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
2. **T. Zhang**, C. Mattson, D. S. Brown. "Imitation Bootstrapped Multi-Agent Reinforcement Learning." Submitted to *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2027.

Workshop papers

3. **T. Zhang**, C. Mattson, D. S. Brown. "Imitation Bootstrapped Multi-Agent Reinforcement Learning." *PhysCollaBot Workshop, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, PA, Sep 2026. **Best Paper Finalist.**

PRESENTATIONS

1. **T. Zhang**, C. Mattson, D. S. Brown. "Imitation Bootstrapped Multi-Agent Reinforcement Learning." Talk and poster, PhysCollaBot Workshop, IROS 2026, Pittsburgh, PA. Sep 2026.
2. **T. Zhang**, A. Voorhis, A. Husband, B. Wright. "Modeling Phenotypic Plasticity with Barbie." Poster, Math4All Conference, Salt Lake City, UT. Apr 6, 2024.

HONORS AND AWARDS

Best Paper Finalist, PhysCollaBot Workshop, IROS 2026	2026
NSF Research Experiences for Undergraduates, BYU Robotics Pioneers REU (NSF Award 2447307)	2026
Best Project Nominee, Scientific Visualization (Prof. Chris Johnson), University of Utah	2025
UROP Research Grant, University of Utah ($2.4K over two semesters)	2024 – 2025
Science Research Initiative (SRI) Summer Research Funding, College of Science, University of Utah	2024
Flagship Scholarship, University of Utah ($28K, merit-based, full undergraduate career)	2023 – 2027
Dean's List, University of Utah (every semester in residence; 5 semesters)	2023 – 2025

TEACHING AND OUTREACH

Mathematics Tutor — Learning Center, University of Utah Aug 2025 – Present

- Tutor NCAA student-athletes in weekly standing sessions on college algebra, quadratic equations, statistics, and calculus.
- Hold appointment-based sessions for students across the university, adapting explanations to each student's course and preparation.

Volunteer Helper — Chip Camp, Brigham Young University Summer 2026

Department of Electrical and Computer Engineering

- Helped run a three-day engineering camp for students aged 12–14; led chemistry demonstrations and helped campers build and launch model rockets.

TECHNICAL SKILLS

Programming: Python, Julia, C++, R, MATLAB

Methods: multi-agent reinforcement learning, imitation learning, belief-space planning, state estimation (Kalman filtering, covariance intersection), diffusion policies, computational simulation

Software: PyTorch, NumPy, ROS, VMAS, HoloOcean, IsaacSim, MuJoCo, ParaView, Git, LaTeX, HPC cluster computing (Utah CHPC)

LANGUAGES

English (native) · Mandarin Chinese (heritage speaker)